

IIT Madras Research Presentation

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Learning Interpretable Models for Autonomous Assessment of Taskable AI Systems

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Research Area & Interesting Problems

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Agent Interrogation Algorithm (AAAI 2021)

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Long Term Research and Teaching Plans

Automated Planning

- To complete any task, we plan using a sequence of actions.

- *Automated Planning*:
  - Has discrete high-level actions.
  - Each high-level action corresponds to a sequence of low-level actions or controls.
  - Given an initial state and a goal, what sequence of actions, called a plan, should an AI system execute to achieve the goal?

Probabilistic Planning Domain Definition Language

```
(:action open-door
   :parameters (?l1)
   :precondition (and
      (has_key)
      (player_at ?l1)
      (door_adjacent ?l1))
   :effect (probabilistic
      0.95 (and (door_open))
      0.05 (and (not (has_key))
                (game-over))
)
```

Precondition: This condition must be true for this action to execute

Effect: This is a set of conditions, one of which becomes true when this action is executed

Probabilities: Each set of effect has an associated probability with which that effect set is executed

Interpretable: Easily Convertible to Natural Language

```
(:action open-door
  :parameters (?l1)
  :precondition (and
     (has_key)
     (player_at ?l1)
     (door_adjacent ?l1))
  :effect (probabilistic
     0.95 (and (door_open))
     0.05 (and (not(has_key))
               (game-over))
)
```

The player can open the door when in location ?l1 if:

- It has the key
- The player is at location ?l1
- The door is adjacent to location ?l1

After executing that capability:

- With 95% probability, the door will open
- With 5% probability, the player will not have the key and the game will be over

Personalized Assessment of AI Systems that can Learn and Plan

- Users would like to give AI systems multiple tasks.
  - How would users know what the AI systems can do?
- AI systems should support third-party assessment.
- The assessment should work with:
  - Adaptive AI systems.
  - Black-Box AI systems.



```
at(p0,cell_6_3)
clear(cell_0_0)
door_at(cell_9_2)
next_to(m0)
alive(m0)
key_at(9_4)
```



[Input]
Concepts that the user understands



Personalized AI-Assessment Module (AAM)



Black-Box AI

Arbitrary internal implementation

Doesn't know user's vocabulary

```
at(p0,cell_6_3)
clear(cell_0_0)
door_at(cell_9_2)
next_to(m0)
alive(m0)
key_at(9_4)
```



[Input]
Concepts that the user understands



Personalized AI-Assessment Module (AAM)

Query

Response





simulator

Black-Box AI

Arbitrary internal implementation

Doesn't know user's vocabulary



[Input]
Concepts that the user understands

[Output]
User-Interpretable description of Black-Box AI's capabilities



Personalized AI-Assessment Module (AAM)

My Research

Query

Response





simulator

Arbitrary internal implementation

Doesn't know user's vocabulary

Black-Box AI

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Deterministic and Stationary Setting

Input

- Predicates (User vocabulary)
  - With their evaluation functions
- List of capabilities.

Output

- PDDL-like description of each capability.



Siddharth Srivastava



Shashank R Marpally

Asking the Right Questions: Learning Interpretable Action Models Through Query Answering

Pulkit Verma, Shashank Rao Marpally, Siddharth Srivastava

In Proceedings of the Thirty-Fifth AAAI Conference on Artificial Intelligence, 2021 (CORE Ranking: A*)

Assumptions

- User's vocabulary matches simulator's vocabulary.
- Black-Box AI provides a list of capabilities.
- Stationary agent model.
- Deterministic environment.
- Fully observable setting.

Exponential Search for Learning Correct Description

- Consider the following 4 predicates/concepts:
  - `(has_key)`
  - `(door_open)`
  - `(door_adjacent ?x)`
  - `(player_at ?x)`
- Consider just one capability: `(open-door ?x)`
- $9^{|C|\times|P|}$ = $9^{1\times4}$=6561 possible models (Assuming deterministic models/ descriptions, i.e., no probabilities).

```
(:action open-door
  :parameters (?l1)
  :precondition (and
    (+/-/∅)(has_key)
    (+/-/∅)(door_open)
    (+/-/∅)(door_adjacent ?l1)
    (+/-/∅)(player_at ?l1))
  :effect (and
    (+/-/∅)(has_key)
    (+/-/∅)(door_open)
    (+/-/∅)(door_adjacent ?l1)
    (+/-/∅)(player_at ?l1))
```

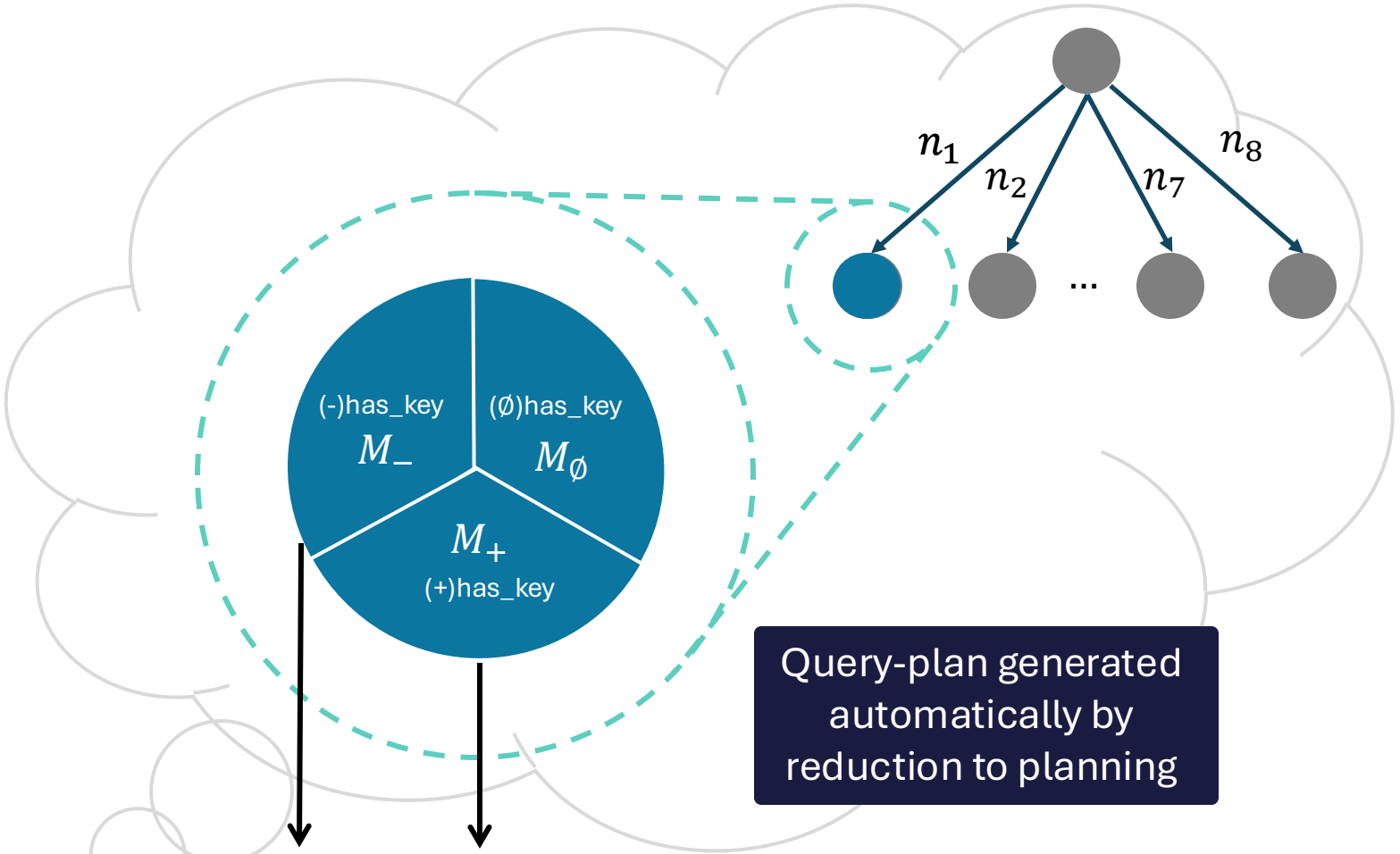
Simple Queries

	Plan Outcome Queries	State Reachability Query
Query	In state s_I, what will happen if you execute the plan $\pi = \langle c_1, \ldots, c_n \rangle$?	Can you go from state s_I to state s_F?
Response	I can execute first ℓ steps of the plan, ending up in state s_F.	Yes / No.

- How to generate the queries?
- How to use the responses to generate models?

We have a reduction that converts this to a planning problem, so it automatically generates queries.

Hierarchical Query Synthesis



Query-plan generated automatically by reduction to planning

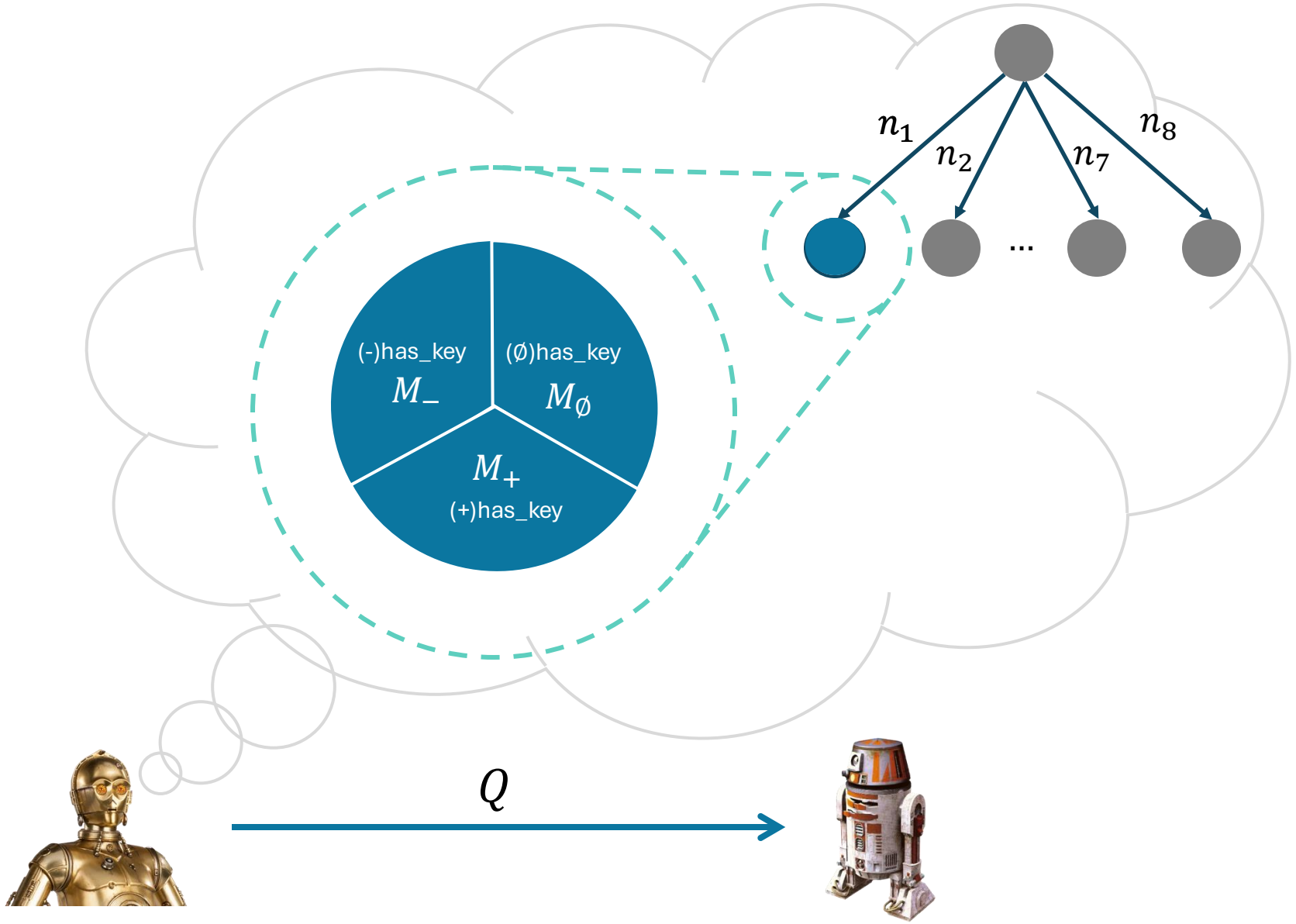
```
(:action open-door
  :parameters (?l1)
  :precondition (and
 n1  (+/-/∅)(has_key)
 n2  (+/-/∅)(door_open)
 n3  (+/-/∅)(door_adjacent ?l1)
 n4  (+/-/∅)(player_at ?l1))
  :effect (and
 n5  (+/-/∅)(has_key)
 n6  (+/-/∅)(door_open)
 n7  (+/-/∅)(door_adjacent ?l1)
 n8  (+/-/∅)(player_at ?l1))
```



Generate a *distinguishing query:* Q such that $Q(M_-) \neq Q(M_+)$

[**Verma**, Marpally, Srivastava; AAAI '21]

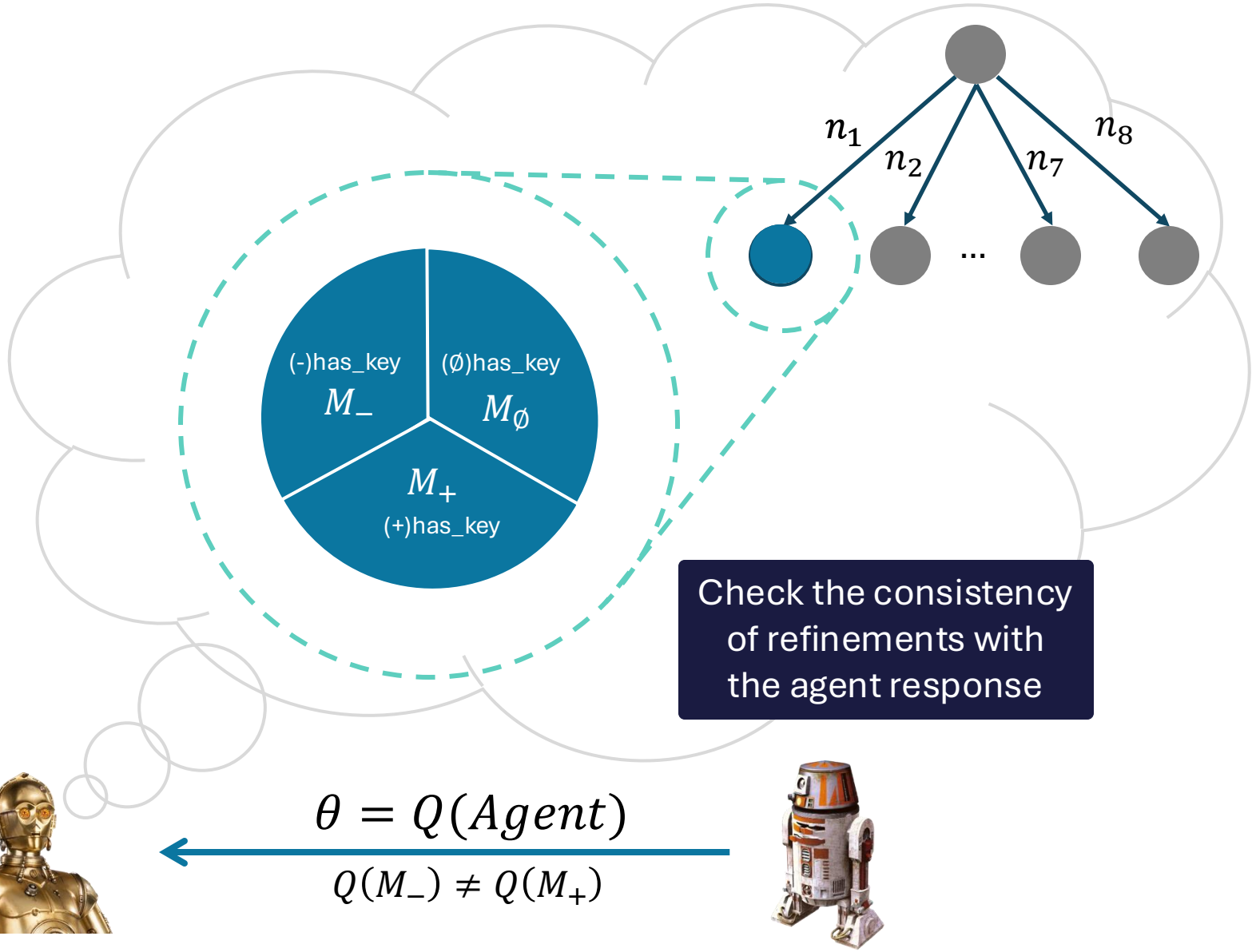
Hierarchical Query Synthesis




```
(:action open-door
  :parameters (?l1)
  :precondition (and
 n1  (+/-/∅)(has_key)
 n2  (+/-/∅)(door_open)
 n3  (+/-/∅)(door_adjacent ?l1)
 n4  (+/-/∅)(player_at ?l1))
  :effect (and
 n5  (+/-/∅)(has_key)
 n6  (+/-/∅)(door_open)
 n7  (+/-/∅)(door_adjacent ?l1)
 n8  (+/-/∅)(player_at ?l1))
```

[**Verma**, Marpally, Srivastava; AAAI '21]

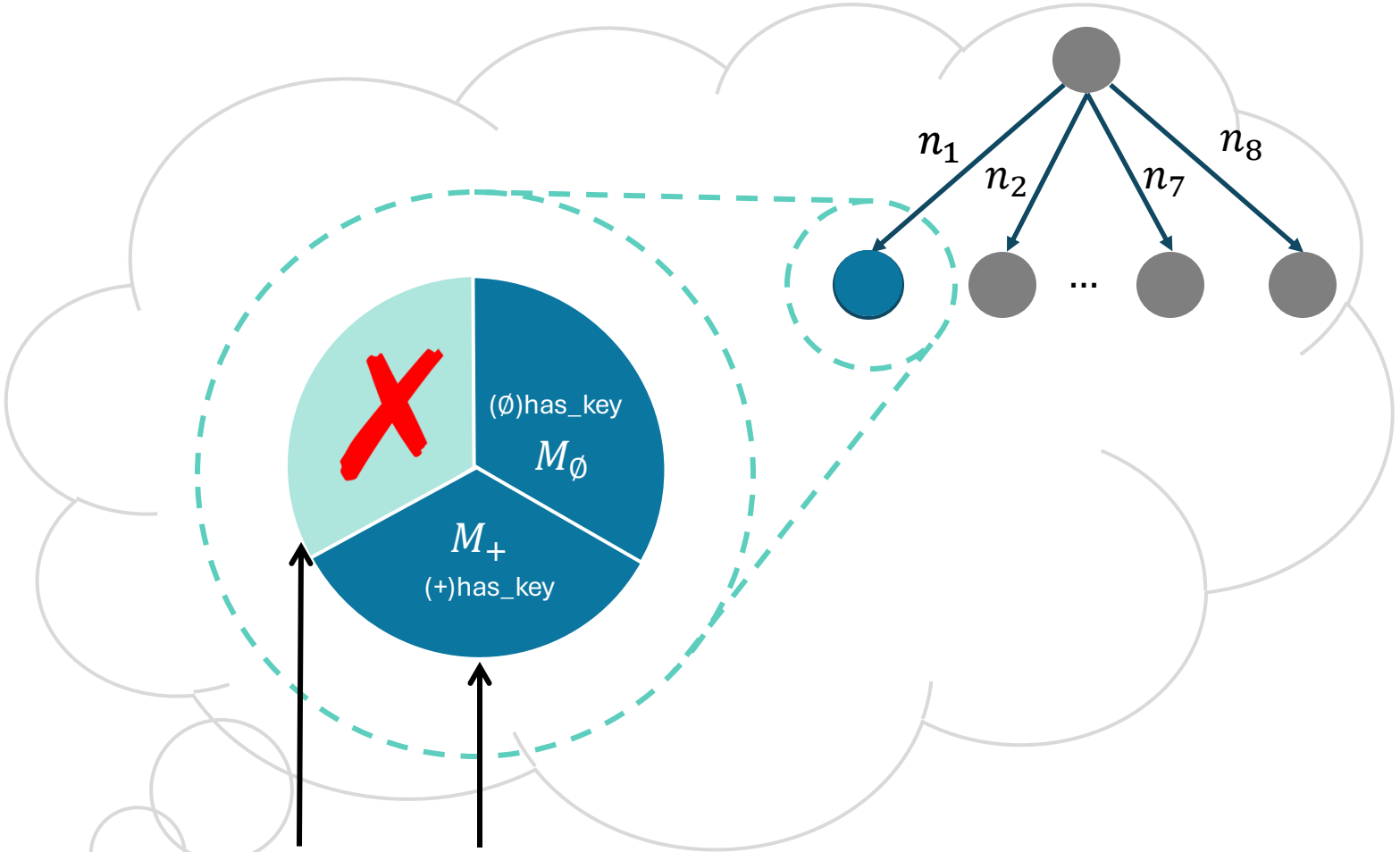
Hierarchical Query Synthesis




```
(:action open-door
  :parameters (?l1)
  :precondition (and
 n1  (+/-/∅)(has_key)
 n2  (+/-/∅)(door_open)
 n3  (+/-/∅)(door_adjacent ?l1)
 n4  (+/-/∅)(player_at ?l1))
  :effect (and
 n5  (+/-/∅)(has_key)
 n6  (+/-/∅)(door_open)
 n7  (+/-/∅)(door_adjacent ?l1)
 n8  (+/-/∅)(player_at ?l1))
```

[**Verma**, Marpally, Srivastava; AAAI '21]

Hierarchical Query Synthesis



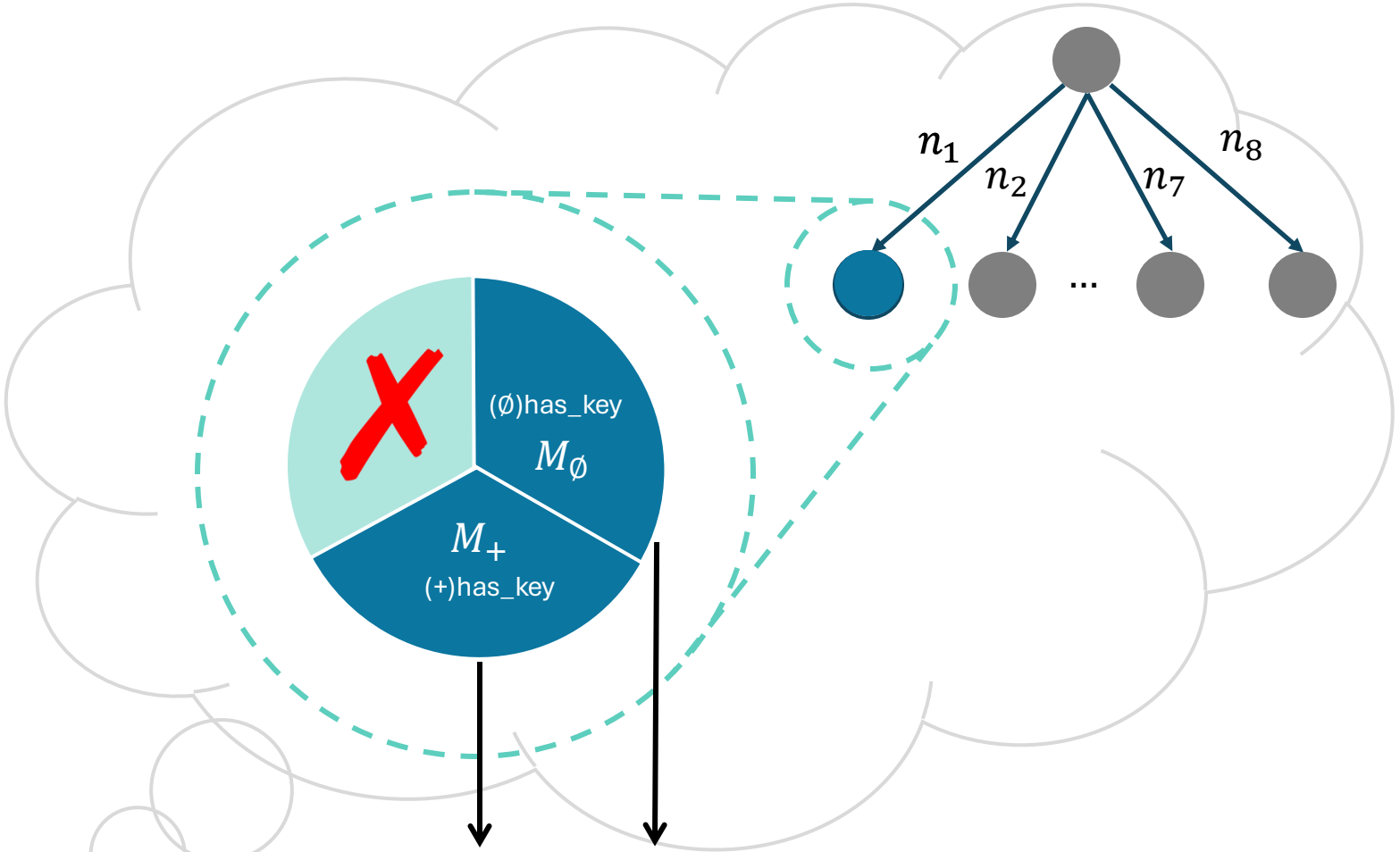

```
(:action open-door
  :parameters (?l1)
  :precondition (and
n1   (+/∅)(has_key)
n2   (+/-/∅)(door_open)
n3   (+/-/∅)(door_adjacent ?l1)
n4   (+/-/∅)(player_at ?l1))
  :effect (and
n5   (+/-/∅)(has_key)
n6   (+/-/∅)(door_open)
n7   (+/-/∅)(door_adjacent ?l1)
n8   (+/-/∅)(player_at ?l1))
```



Reject abstract model(s) that are not consistent with the agent

[**Verma**, Marpally, Srivastava; AAAI '21]

Hierarchical Query Synthesis



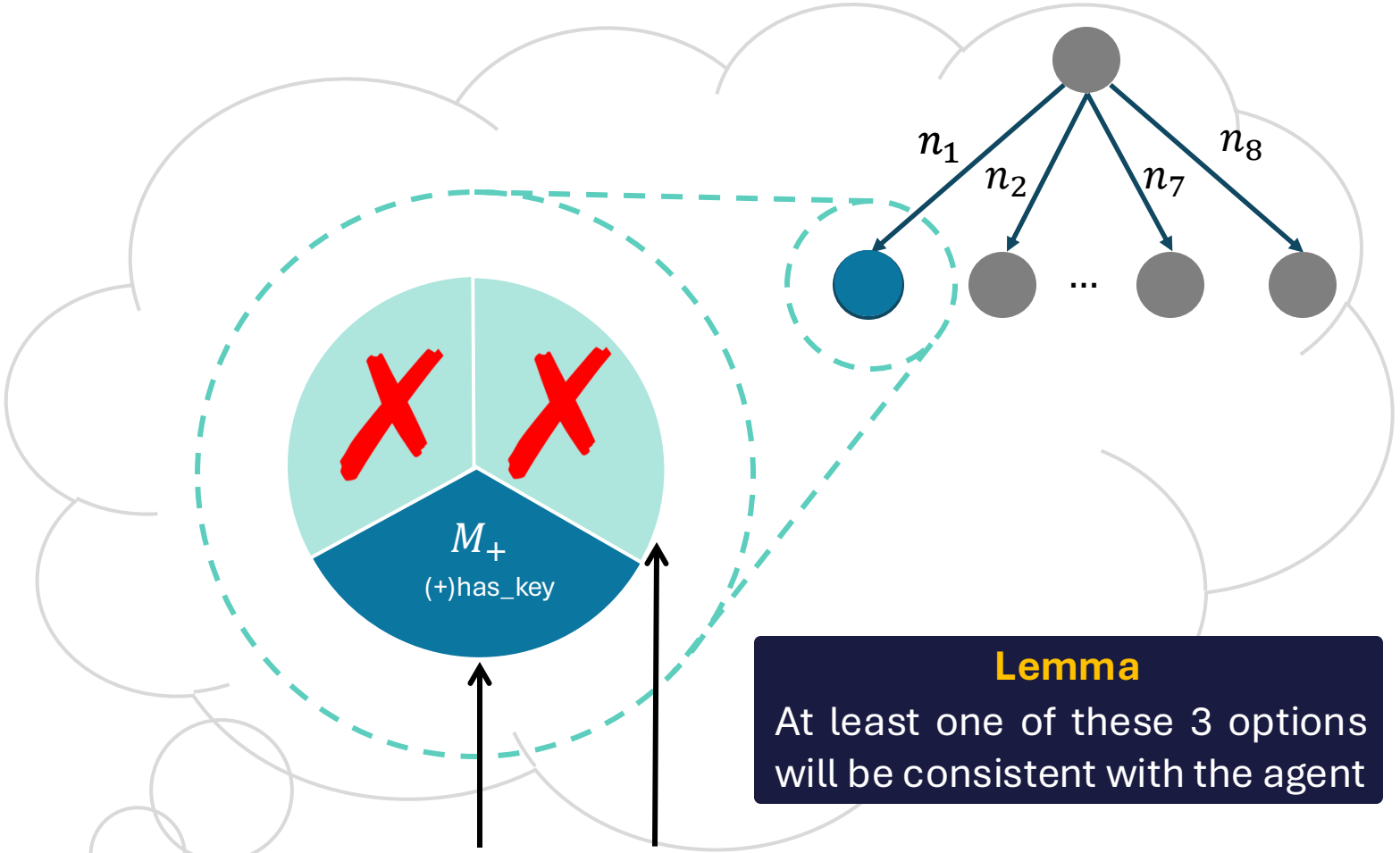
Generate a distinguishing query
for these two abstract models



```
(:action open-door
  :parameters (?l1)
  :precondition (and
n1   (+/Ø)(has_key)
n2   (+/-/Ø)(door_open)
n3   (+/-/Ø)(door_adjacent ?l1)
n4   (+/-/Ø)(player_at ?l1))
  :effect (and
n5   (+/-/Ø)(has_key)
n6   (+/-/Ø)(door_open)
n7   (+/-/Ø)(door_adjacent ?l1)
n8   (+/-/Ø)(player_at ?l1))
```

[**Verma**, Marpally, Srivastava; AAAI '21]

Hierarchical Query Synthesis



Lemma
At least one of these 3 options will be consistent with the agent

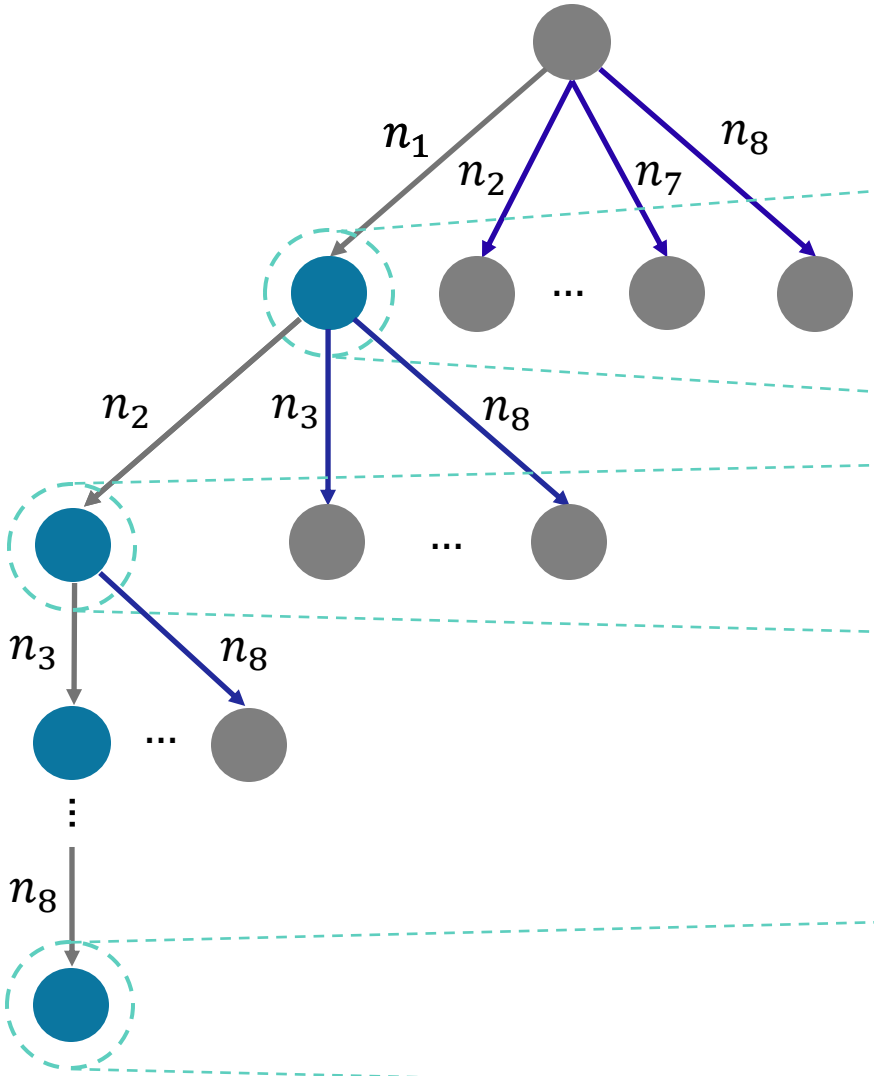
```
(:action open-door
  :parameters (?l1)
  :precondition (and
 n1      (+)(has_key)
 n2   (+/-/∅)(door_open)
 n3   (+/-/∅)(door_adjacent ?l1)
 n4   (+/-/∅)(player_at ?l1))
  :effect (and
 n5   (+/-/∅)(has_key)
 n6   (+/-/∅)(door_open)
 n7   (+/-/∅)(door_adjacent ?l1)
 n8   (+/-/∅)(player_at ?l1))
```

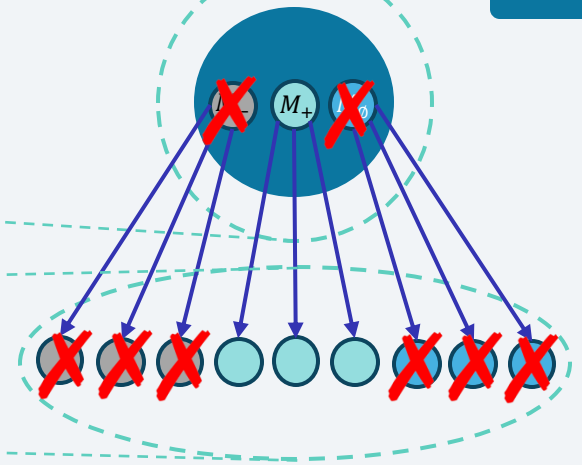


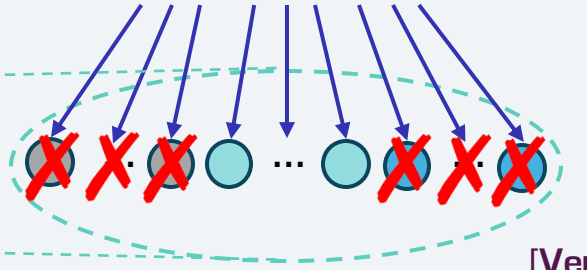
Reject the abstract model
that is not consistent with the agent

[**Verma**, Marpally, Srivastava; AAAI '21]

Hierarchical Query Synthesis







Key feature of the algorithm

Whenever we prune an abstract model, we prune a large number of concrete models.

Active Learning

[**Verma**, Marpally, Srivastava; AAAI '21]

AAM learns Accurate Model with fewer Queries

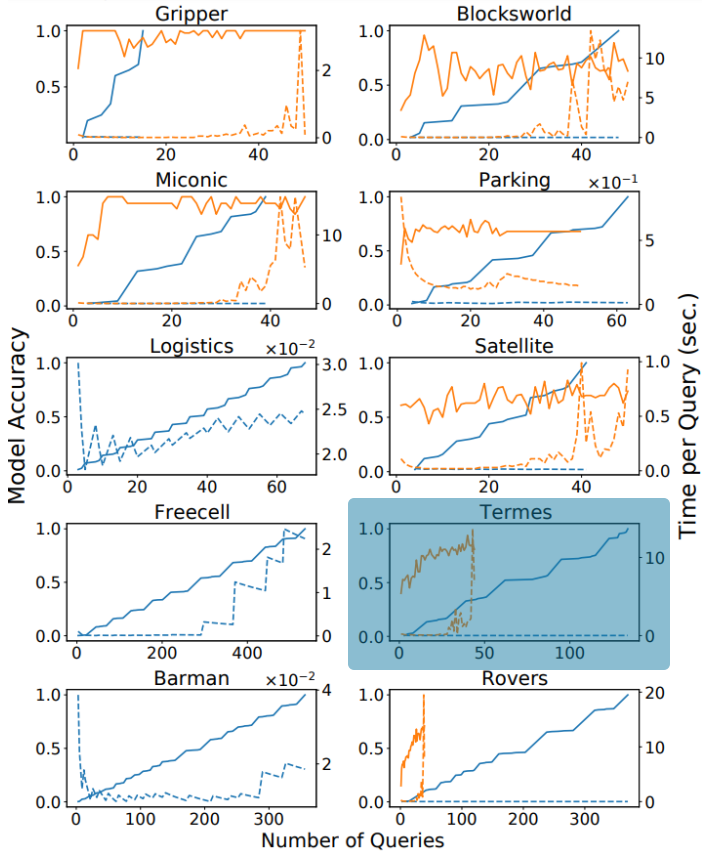
- Asses by learning the model and compare with ground truth.
- Baseline†: A passive learner (FAMA) that observes agent behavior

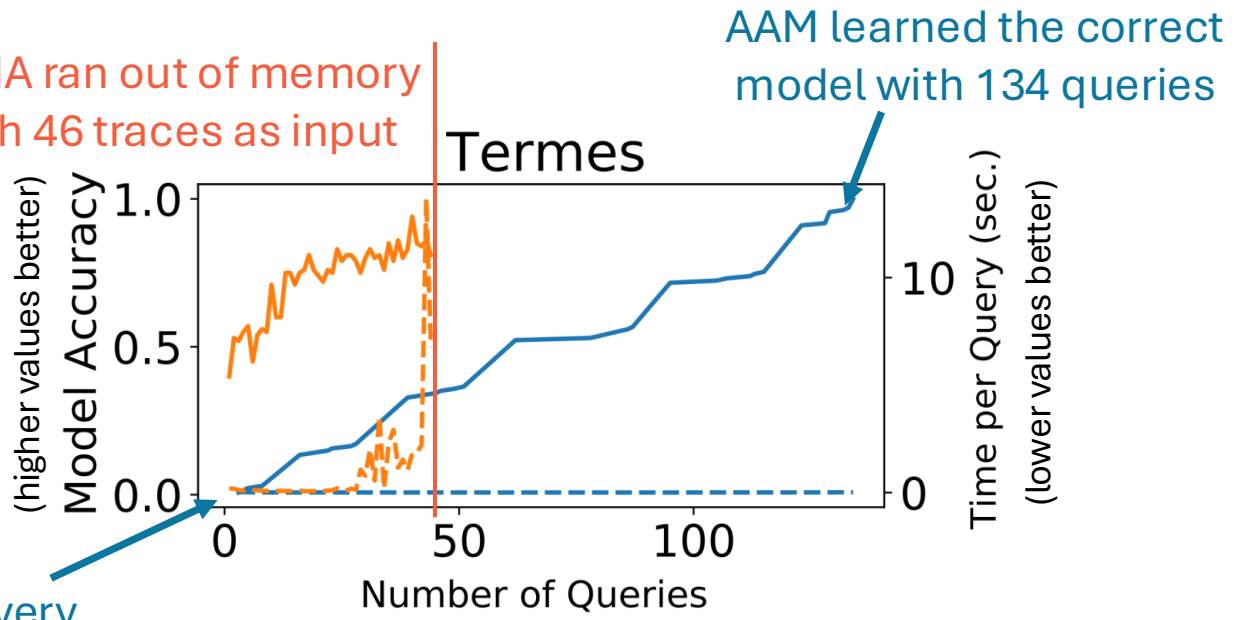


Random deterministic planning agent from IPC

Accuracy: AAM FAMA

Time: AAM FAMA





[**Verma**, Marpally, Srivastava; AAAI '21]

†Aineto, D.; Celorrio, S. J.; and Onaindia, E. 2019. *Learning Action Models With Minimal Observability*. Artificial Intelligence 275: 104–137.

AAM learns Accurate Deterministic Models

- Theorem (*termination*) : The algorithm terminates after a finite number of iterations.

- Theorem (*soundness*): The resulting (set of) model(s) is(are) functionally equivalent to the ground truth model.

Theorem 3 in Paper

Stochastic and Stationary Setting

Input

- Predicates (User vocabulary)
  - With their evaluation functions
- List of capabilities.

Output

- PPDDL-like description of each capability.

[**Verma**, Karia, Srivastava; NeurIPS '23]

Assumptions

- User's vocabulary matches simulator's vocabulary.
- Black-Box AI provides a list of capabilities.
- Stationary agent model.
- ~~Deterministic~~ Stochastic environment.
- Fully observable setting.

Discovering Capabilities

Input

- Predicates (User vocabulary)
  - With their evaluation functions
- Samplers: high-level state to low-level state.
- Low-level state transitions.



Output

- List of capabilities.
- PDDL-like description of each capability.

[**Verma**, Marpally, Srivastava; KR '22]

Assumptions

- ~~User's vocabulary matches simulator's vocabulary~~.
- Black-Box AI provides a list of ~~capabilities~~. transitions.
- Stationary agent model.
- Deterministic environment.
- Fully observable setting.

Differential Assessment

Input

- Initial model of the AI system.
  - Predicates (User vocabulary)
    - With their evaluation functions
  - List of capabilities.
- Observations of AI system working in the environment.

Output

- Updated PDDL-like description of each capability.

[Nayyar*, **Verma***, Srivastava; AAAI '22]

Assumptions

- User's vocabulary matches simulator's vocabulary.
- Black-Box AI provides a list of capabilities.
- Adaptive
- ~~Stationary~~ agent model.
- Deterministic environment.
- Fully observable setting.

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Agent Interrogation Algorithm (AAAI 2021)

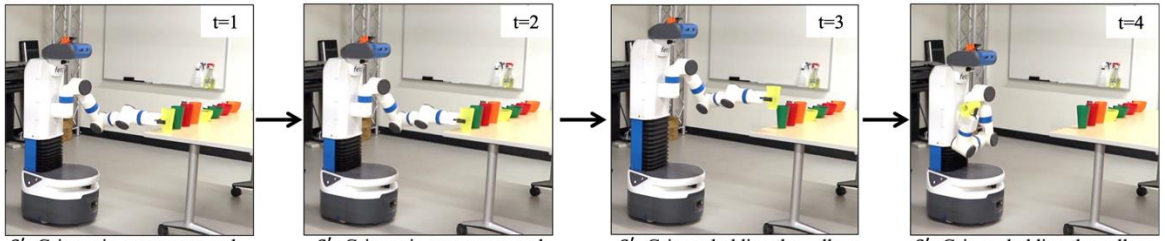
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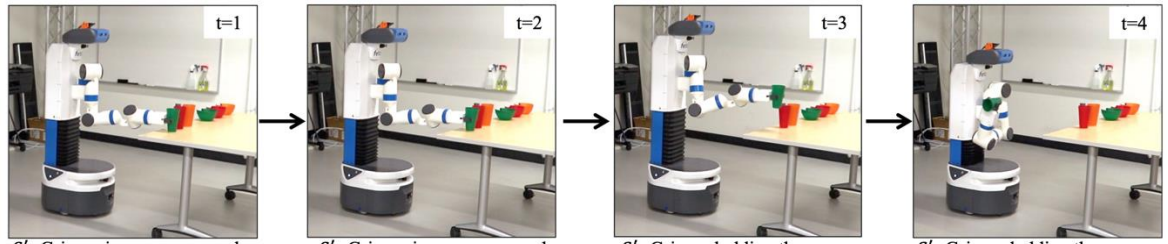
Autonomous Robot Skill Acquisition

Yellow Cup



S'_1: Gripper in grasp pose and not holding the yellow cup.
S'_2: Gripper in grasp pose and holding the yellow cup.
S'_3: Gripper holding the yellow cup, which is not on table.
S'_4: Gripper holding the yellow cup. Robot's arm is tucked.

Green Cup



S'_1: Gripper in grasp pose and not holding the green cup.
S'_2: Gripper in grasp pose and holding the green cup.
S'_3: Gripper holding the green cup, which is not on table.
S'_4: Gripper holding the green cup. Robot's arm is tucked.

First-order State

- S_1 : {(table-can1 ?table ?can), (can-gripper1 ?can ?gripper), (base-gripper0 ?base ?gripper), (base-table1 ?base ?table)}.
- S_2 : {(table-can1 ?table ?can), (can-gripper2 ?can ?gripper), (base-gripper0 ?base ?gripper), (base-table1 ?base ?table)}.
- S_3 : {(table-can0 ?table ?can), (can-gripper2 ?can ?gripper), (base-gripper0 ?base ?gripper), (base-table1 ?base ?table)}.
- S_4 : {(table-can0 ?table ?can), (can-gripper2 ?can ?gripper), (base-gripper1 ?base ?gripper), (base-table1 ?base ?table)}.

Learned Skills

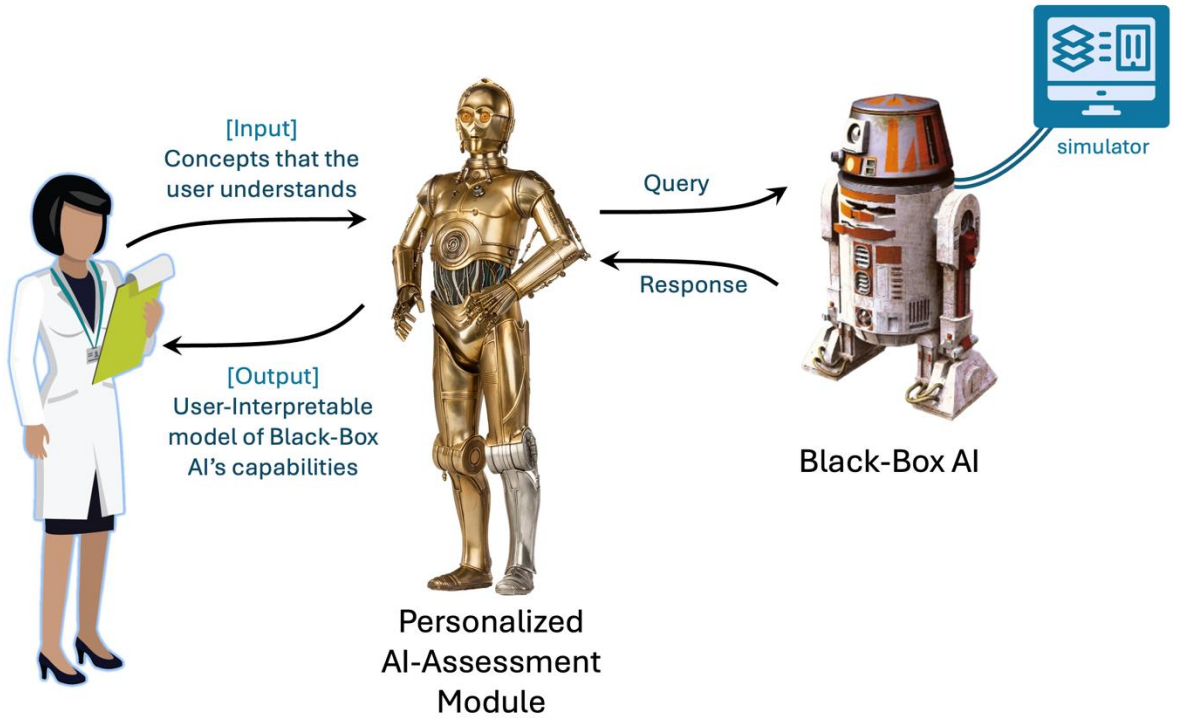
- $C_{12} = \langle {}^{+}C_{12} = \{$(can-gripper2 ?can ?gripper)$\}, {}^{-}C_{12} = \{$(can-gripper1 ?can ?gripper)$\}\rangle$.
- $C_{23} = \langle {}^{+}C_{23} = \{$(table-can0 ?table ?can)$\}, {}^{-}C_{23} = \{$(table-can1 ?table ?can)$\}\rangle$.
- $C_{34} = \langle {}^{+}C_{34} = \{$(base-gripper1 ?base ?gripper)$\}, {}^{-}C_{34} = \{$(base-gripper0 ?base ?gripper)$\}\rangle$.

- Currently* depends on manual selection of tasks.
- Goal is to make the task selection autonomous.
  - Using active learning techniques.

*N. Shah, J. Nagpal, **P. Verma**, S. Srivastava. From Reals to Logic and Back: Inventing Symbolic Vocabularies, Actions, and Models for Planning from Raw Data. ArXiv 2402.11871 , 2024

Interpretable Representations

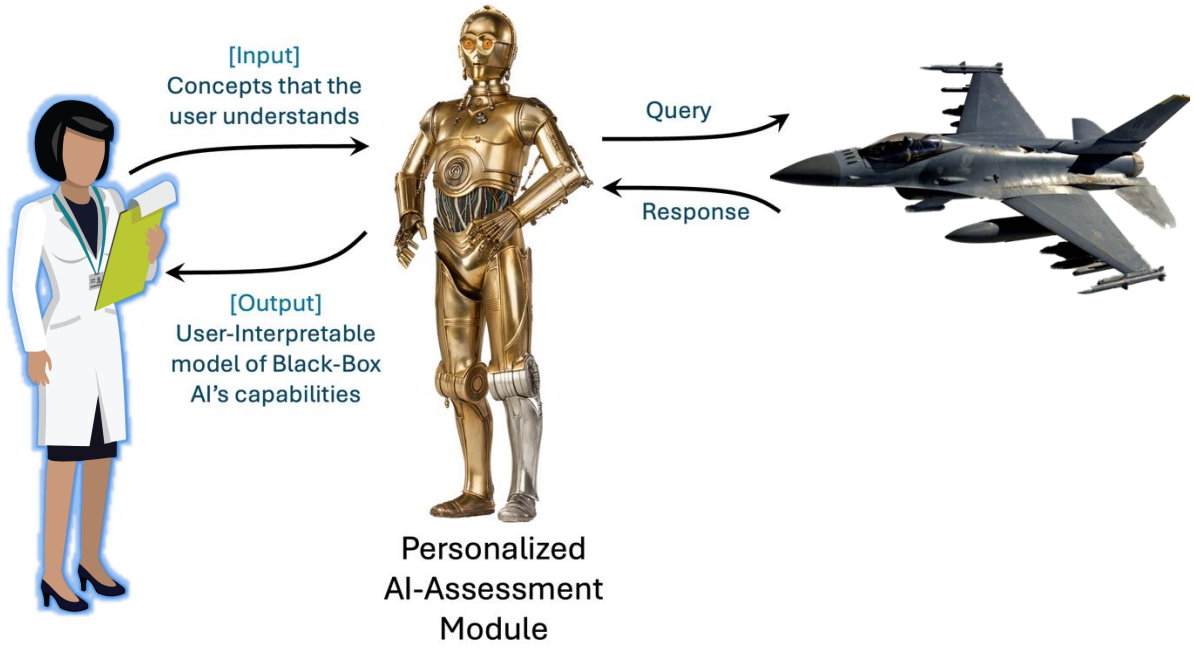
- Representations beyond PDDL.
  - LTL – Linear Temporal Logic
  - STL- Signal Temporal Logic
  - CDL- Crow Definition Language
- How interpretable each representation is?



Real World Setting

- Flying Extended Trail
- Human must collaborate with the AI-powered aircraft.

- How can we explain the aircraft's policy to the human in an interpretable way?



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Contributions

AI Assessment

[**Verma**, Marpally, Srivastava; AAAI '21]

[Nayyar*, **Verma***, Srivastava; AAAI '22]

[**Verma**, Marpally, Srivastava; KR '22]

[**Verma**, Karia, Srivastava; NeurIPS '23]

[**Verma***, Karia*, Vipat, Gupta, Srivastava; GenPlan '23]

[**Verma**, Srivastava; AAAI '24 SS]

[Karia, Dobhal, Bramblett, **Verma**, Srivastava; AAAI '24 SS]

Generalization in Planning

[Karia*, **Verma***, Speranzon, Srivastava; ICAPS '24]

[Shah, Nagpal, **Verma**, Srivastava; Preprint]

[Chen, Verma, Srivastava, Katz, Thiebaux; PRL@AAAI' 25]

Causal Accuracy of Symbolic Models

[**Verma**, Srivastava; GenPlan '21]

[**Verma**, Srivastava; Tech Report, 2024]

Explainable Robot Planning

[Shah*, **Verma***, Angle, Srivastava; AAMAS '22]

[Dobhal, Nagpal, Karia, **Verma**, Nayyar, Shah, Srivastava; EAAI @ AAAI '25]

[Dadvar, Majd, Oikonomou, **Verma**, Fainekos, Srivastava; In Submission]

Contributions

Conferences

- ***Pulkit Verma***, Shashank Rao Marpally, and Siddharth Srivastava. *Asking the Right Questions: Learning Interpretable Action Models through Query Answering*. In AAAI 2021. A*
- Rashmeet Kaur Nayyar*, ***Pulkit Verma****, and Siddharth Srivastava. *Differential Assessment of Black-Box AI Agents*. In AAAI 2022. A*
- ***Pulkit Verma***, Shashank Rao Marpally, and Siddharth Srivastava. *Discovering User-Interpretable Capabilities of Black-Box Planning Agents*. In KR 2022. A*
- Naman Shah*, ***Pulkit Verma****, Trevor Angle, and Siddharth Srivastava. *JEDAI: A System for Skill-Aligned Explainable Robot Planning*. In AAMAS 2022 (Demonstration Track). A*

  🏆 Winner of Best Demo Award
- Yizhong Wang et al. *Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ Tasks*. In EMNLP 2022. A
- ***Pulkit Verma***, Rushang Karia, and Siddharth Srivastava. *Autonomous Capability Assessment of Sequential Decision-Making Systems in Stochastic Settings*. In NeurIPS 2023. A*
- Rushang Karia*, ***Pulkit Verma****, Alberto Speranzon, and Siddharth Srivastava. *Epistemic Exploration for Generalizable Planning and Learning in Non-Stationary Settings*. In ICAPS 2024. A*
- Rushang Karia*, Jayesh Nagpal*, Daksh Dobhal*, ***Pulkit Verma***, Rashmeet Kaur Nayyar, Naman Shah, and Siddharth Srivastava. *Using Explainable AI and Hierarchical Planning for Outreach with Robots*. In EAAI 2025 (Proceedings of AAAI'25).

**Equal Contribution*

Workshops, Symposia, Preprints

- ***Pulkit Verma*** and Siddharth Srivastava. *Learning Generalized Models by Interrogating Black-Box Autonomous Agents*. In AAAI 2020 GenPlan.
- ***Pulkit Verma*** and Siddharth Srivastava. *Learning Causal Models of Autonomous Agents using Interventions*. In IJCAI 2021 GenPlan.
- ***Pulkit Verma***, Shashank Rao Marpally, and Siddharth Srivastava. *Learning User-Interpretable Descriptions of Black-Box AI System Capabilities*. In ICAPS 2021 KEPS.
- ***Pulkit Verma****, Rushang Karia*, Gaurav Vipat, Anmol Gupta, and Siddharth Srivastava. *Learning AI-System Capabilities under Stochasticity*. In NeurIPS 2023 GenPlan.
- ***Pulkit Verma*** and Siddharth Srivastava. *User-Aligned Autonomous Capability Assessment of Black-Box AI Systems*. In AIA 2024.
- Rushang Karia, Daksh Dobhal, Daniel Bramblett, ***Pulkit Verma***, and Siddharth Srivastava. *Can LLMs translate SATisfactorily? Assessing LLMs in Generating and Interpreting Formal Specifications*. In AIA 2024.
- Dillon Z. Chen, ***Pulkit Verma***, Siddharth Srivastava, Michael Katz, and Sylvie Thiébaux. *AI Planning: A Primer and Survey (Preliminary Report)*. In AAAI 2025 PRL.
- Mehdi Dadvar, Keyvan Majd, Elena Oikonomou, ***Pulkit Verma***, Georgios Fainekos, and Siddharth Srivastava. *Joint Communication and Motion Planning for Cobots in Real-World Contexts*. In Submission.
- Naman Shah, Jayesh Nagpal, ***Pulkit Verma***, and Siddharth Srivastava. *From Reals to Logic and Back: Inventing Symbolic Vocabularies, Actions, and Models for Planning from Raw Data*. Preprint.



Rushang Karia



Shashank Marpally



Rashmeet Nayyar



Alberto Speranzon



Siddharth Srivastava







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Long Term Research and Teaching Plans

Lifelong Learning with Human-in-the Loop



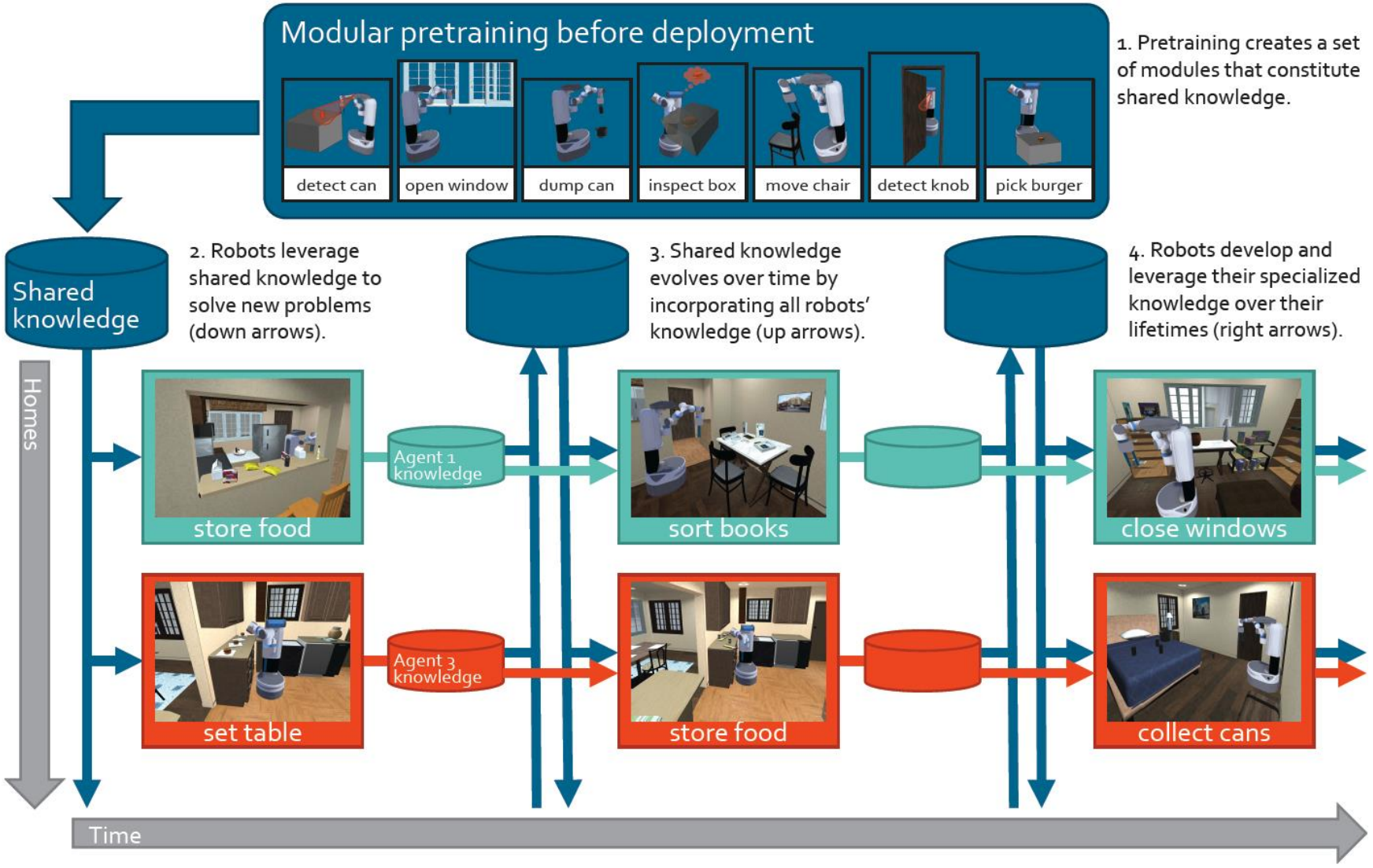


Image Courtesy: Jorge Mendez-Mendez, IRL Lab, MIT

Warehouse Robotics

- The increasing number of Indian robotics companies makes research accessible with cheaper robots.
- Possible work on removing the need for warehouse robots to follow fixed paths.
- Can start research on Future of work where humans and robots work in same space.

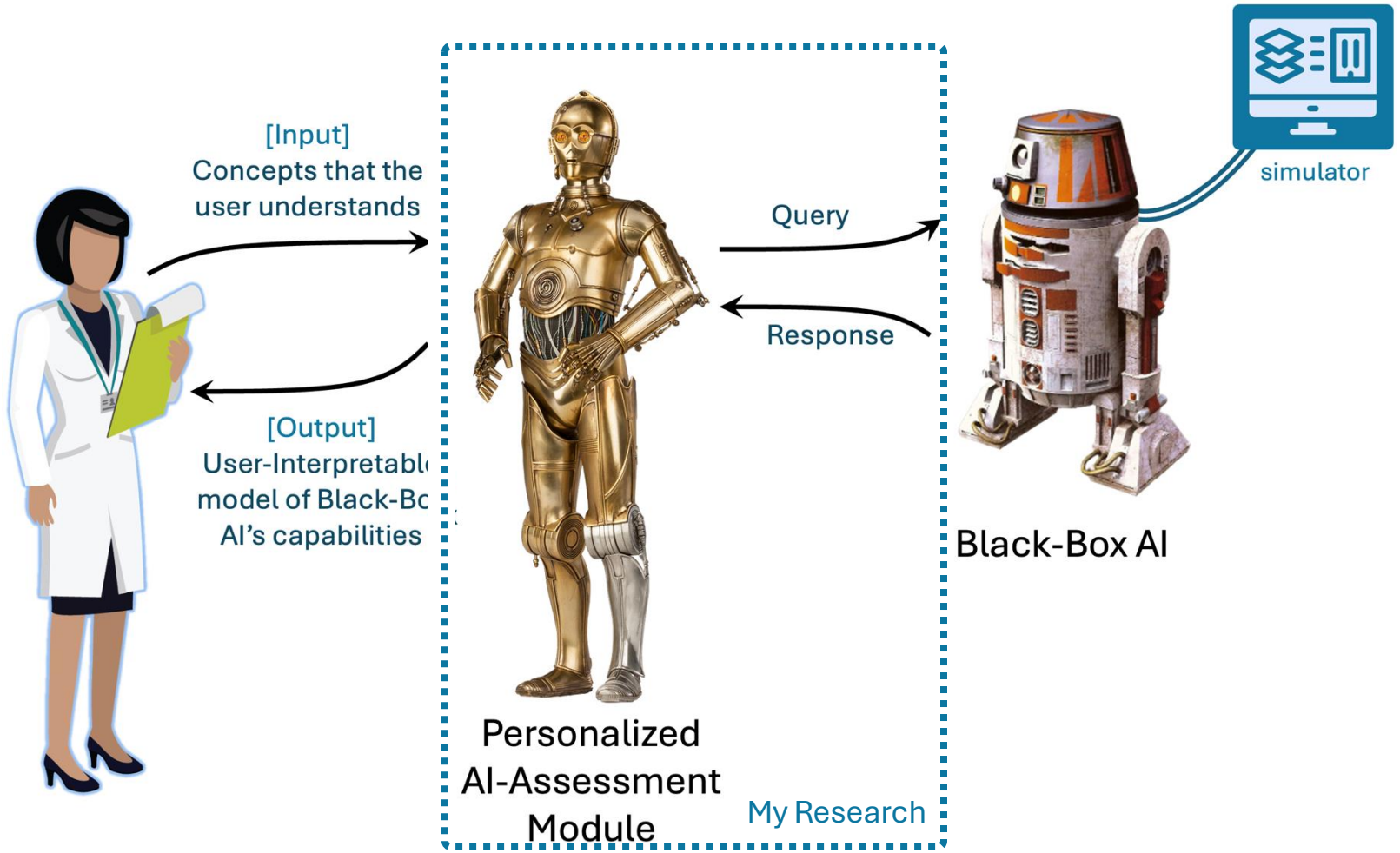
ADDVERB
Human | Robot | Possibilities

Noida-based
Warehouse Robot Manufacturer



Learning Interpretable Models for Autonomous Assessment of Taskable AI Systems

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Questions?